

Simultaneous Equations: Finding the Secret Crossing

The Secret Crossing Point Where Two Lines Meet

Lesson Handout • Instructors: Rahul & Amit

1. The Treasure Map: Our Grid as a Neighborhood

Imagine the coordinate grid as a map of your local neighborhood! Every point on the grid corresponds to a unique house, landmark, or location.

- **Home Base (Origin):** We always start at zero, represented as $(0, 0)$.
- **Moving Forward (Right):** Count steps horizontally along the bottom axis (x -axis).
- **Moving Upward:** Count steps vertically along the side axis (y -axis).

Key Concept

Every crossing point on the grid is a special address. Today, our goal is to find the most secret address of all — where two rules cross paths!

2. Lines are Rules: Real-World Scenarios

A straight line on a graph represents a mathematical **rule** or equation. Every point lying on that line is a valid solution to that rule.

Scenario A: The Sweet Shop Puzzle (Priya)

Priya has ₹10 to spend on Jalebis (x) and Laddoos (y). The rule line connects all exact combinations to spend ₹10:

Jalebis (x)	Laddoos (y)	Coordinate Point (x, y)	Total Spent
0	5	$(0, 5)$	0 Jalebis + 5 Laddoos = ₹10
2	4	$(2, 4)$	2 Jalebis + 4 Laddoos = ₹10
4	3	$(4, 3)$	4 Jalebis + 3 Laddoos = ₹10
6	2	$(6, 2)$	6 Jalebis + 2 Laddoos = ₹10
8	1	$(8, 1)$	8 Jalebis + 1 Laddoo = ₹10

Scenario B: Market Shopping (Rahul)

Rahul also has ₹10 to spend on Samosas (x) and Kites (y):

- $(0, 2) \rightarrow 0$ Samosas + 2 Kites = ₹10
- $(1, 1) \rightarrow 1$ Samosa + 1 Kite = ₹10
- $(2, 0) \rightarrow 2$ Samosas + 0 Kites = ₹10

Scenario C: Stationery Shop Challenge (Kavya)

Kavya has ₹10. Pencils cost ₹2 each and Erasers cost ₹1 each (Equation: $2x + y = 10$):

- Points on Kavya's line: $(0, 10), (1, 8), (2, 6), (3, 4), (4, 2), (5, 0)$

3. The Secret Crossing Point & Summary

When we plot two different rule lines on the exact same grid, the point where they cross is called the **Secret Crossing Point** (Intersection Point).

Why is the Crossing Point Special?

It is the **one unique point** where BOTH rules are true at the exact same time! It gives the simultaneous solution that satisfies both conditions together.

What Did We Learn Today?

- **The Grid is a Map:** We navigate forward (right) and up using coordinates (x, y) .
- **Lines are Rules:** Each line connects all possible choices that obey a specific rule or equation.
- **The Crossing Point:** The intersection point of two lines provides the secret answer satisfying both equations simultaneously.

Student Practice Sheet: Simultaneous Equations

Solve the following problems by finding the secret crossing point. Show your work!

Question 1: Cricket Scores

Rahul and Amit are playing cricket. Together, their run combinations form two rules:

- Rule 1: Rahul's runs (x) + Amit's runs (y) = 10
- Rule 2: Amit's runs are 2 more than Rahul's runs ($y = x + 2$)

If Rahul scores 4 runs and Amit scores 6 runs, verify if $(4, 6)$ is the secret crossing point. Write down your explanation and verify coordinates on the grid.

Question 2: Parallel Rules Challenge

Two mathematical rules create two lines on a coordinate grid:

- Line 1: $x + y = 10$
- Line 2: $x + y = 6$

Will these two lines ever cross on the grid? Why or why not? Explain what this means regarding a shared secret answer.

Question 3: Finding the Sweet & Snack Crossing

Priya spends money according to the rule $x + 2y = 10$ (where x is Jalebis and y is Laddoos). Rahul spends money according to the rule $x - y = 1$.

Find the secret crossing point (x, y) where both Priya's rule and Rahul's rule are true at the same time.

Question 4: Graphing & Intersection Practice

Graph the following two equations on a grid and find their secret crossing point:

- Equation 1: $y = 2x$
- Equation 2: $x + y = 6$

State the exact coordinates (x, y) of the intersection point.

Question 5: Real-World Word Problem

A school stationery shop sells notebooks (n) for ₹5 each and pens (p) for ₹2 each. Rohan spent ₹20 in total ($5n + 2p = 20$). He bought a total of 7 items ($n + p = 7$).

Construct a grid or table of values to find how many notebooks and pens Rohan bought.